

regarding the specific skills and contexts required for AI integration. Additionally, the GENOE index considers the complex interplay of tasks within occupations and integrating the ethical and social contexts organically.

Another contribution of our paper is the novel use of large language models to generate synthetic assessments that aim to replicate and extend expert assessments. Many studies rely on expert surveys or interviews to build analytical tools like indexes, including in the labor automation area (Frey and Osborne, 2017; Felten *et al.*, 2021; Tolan *et al.*, 2021). Synthetic AI surveys offer several advantages. They can efficiently process and synthesize vast amounts of information from diverse sources very rapidly, enabling a larger volume of questions to be addressed in a shorter time frame. Additionally, synthetic AI surveys can be easily scaled to include extensive sample sizes or repeated iterations, allowing greater granularity of the collected data.

While recent research has begun to explore the use of AI in workforce analysis, our methodology offers a novel approach in this evolving field. For example, Eloundou *et al.* (2024) combined human expert insights with AI-generated assessments to evaluate AI’s potential for enhancing task efficiency across occupations. Our work is a step forward in this approach by exclusively and systematically using ChatGPT to build the entire index through multiple iterations and varying model parameters.

To validate the accuracy of this methodology, we replicated the well-known exposure index of Felten *et al.* (2021), replacing mTurk surveys with synthetic AI surveys. Our approach successfully replicated Felten’s AI Occupational Exposure Index (AIOE) with a correlation of 0.99, suggesting our methodology is robust and reliable for these types of tasks and providing evidence of another area where AI-driven automation is highly feasible.

The remainder of this paper is organized as follows. After this introduction, Section 2 details our data sources and methodology used to build the GENOE index. Section 3 presents the GENOE index, including its key findings and implications. This section also includes several robustness checks to validate our results and replicates other human-like expert assessment using our novel synthetic survey methodology. Section 4 explores how the index can be used in practical applications, analyzing occupational exposure in the U.S. and Mexico labor market. Finally, Section 5 discusses some considerations and limitations of our research, while Section 6 concludes with implications and directions for future research.

2 DATA AND METHODOLOGY

We draw data from the O*NET database, which contains detailed descriptions and attributes for 1,016 occupations in the U.S. labor market under the O*NET-SOC 2019 classification. Occupational characterizations are built and continuously updated from surveys and input from multiple sources, including job analysts, incumbent workers, and industry experts. O*NET attributes—such as abilities, skills, knowledge, and tasks—are divided into six major domains: Worker Characteristics, Worker Requirements, Experience Requirements, Occupational Requirements, Work-

veys. To ensure compatibility with standard occupational classifications, we translated these estimations to the SOC-18 classification using O*NET’s crosswalks for a total of 759 occupations.

To enhance our understanding of the GENOE’s effectiveness and to validate our methodology, we conducted several additional exercises. Firstly, we measured task-level exposure by assessing the proportion of tasks at risk within each occupation, allowing us to identify roles where AI might significantly impact certain tasks without necessarily replacing the entire job. We also evaluated tasks in isolation and compared the results with our holistic occupation-level approach to understand the value of considering tasks within their broader occupational context. Additionally, we tested the effect of withholding the occupation name from the model to gauge the importance of occupational context in the AI’s assessment. To incorporate broader societal considerations, we explicitly prompted the model to consider ethical and social factors in its assessments.

Furthermore, we conducted follow-up surveys, asking the AI to explain its reasoning behind the exposure estimates, effectively “opening the black box” of its decision-making process. These additional analyses serve multiple purposes: they validate our methodology by providing alternative perspectives on AI exposure, offer insights into the factors influencing AI’s potential impact on various occupations, and help us understand the decision-making process of the AI model. This approach not only strengthens the robustness of our GENOE index but also potentially reveals any biases or limitations in the AI’s assessments, providing a more reliable picture of occupational exposure to AI.

To further validate our approach, we applied our synthetic survey methodology to replicate the Felten *et al.* (2021) index of occupational exposure. In this exercise, we substituted the original human expert surveys with synthetic surveys conducted using ChatGPT4o. The primary objective was to assess the capability of our AI-driven methodology to reproduce results comparable to those obtained through traditional human expert assessments. This replication serves as a crucial test of our approach’s reliability and accuracy, offering insights into the potential of large language models to emulate human expertise in evaluating occupational exposure to AI.

3 RESULTS

3.1 Baseline Estimations

Figure 1 presents a comprehensive view of AI exposure across 759 occupations, illustrating the average, median, 25th percentile, and 75th percentile for the three defined time horizons. The analysis reveals a progressive increase in average AI exposure: 0.28 within one year, rising to 0.38 over five years, and reaching 0.44 over a decade. This pattern suggests that as AI technologies advance and mature, their potential to reshape various professions grows significantly.⁵

⁵Complete descriptive statistics can be found in Appendix Table A.1.